SSC Maharashtra Board — Geometry (Mathematics Part II)

March 2026 — Model Solutions

Time: 2 Hours | Max. Marks: 40

Q.1

Q.1 (A) Choose the correct alternative from given: [4 Marks]

(1) Out of the following which is a Pythagorean triplet?

(A) (1, 5, 10) (B) (3, 4, 5) (C) (2, 2, 2) (D) (5, 5, 2)

Check option (B): $3^2 + 4^2 = 9 + 16 = 25$ and $5^2 = 25$.
$\therefore 3^2 + 4^2 = 5^2$
$\therefore$ (3, 4, 5) is a Pythagorean triplet.

$$\boxed{\text{Ans: (B) } (3, 4, 5)}$$

(2) Two circles having radii 4 cm and 5 cm touch each other internally. Find the distance between their centres.

(A) 4 cm (B) 5 cm (C) 1 cm (D) 9 cm

When two circles touch internally, the distance between their centres is equal to the difference of their radii.
$\therefore$ Distance $= 5 - 4 = 1$ cm

$$\boxed{\text{Ans: (C) 1 cm}}$$

(3) Out of the following points ... lies to the right of the origin on X-axis.

(A) $(-2, 0)$ (B) $(0, 2)$ (C) $(2, 3)$ (D) $(2, 0)$

A point lies on the X-axis if its y-coordinate is 0.
It lies to the right of origin if its x-coordinate is positive.
$(2, 0)$ has $y = 0$ and $x = 2 > 0$.

$$\boxed{\text{Ans: (D) } (2, 0)}$$

(4) If the side of the cube is 5 cm, then the volume of that cube is ...

(A) 10 cm^3 (B) 100 cm^3 (C) 125 cm^3 (D) 25 cm^3

Volume of cube $= (\text{side})^3 = (5)^3 = 125$ cm^3

$$\boxed{\text{Ans: (C) 125 cm}^3}$$

Q.1 (B) Solve the following sub-questions: [4 Marks]

(1) The ratio of corresponding sides of similar triangles is 3 : 5, then find the ratio of their areas.

If two triangles are similar and the ratio of their corresponding sides is $\frac{3}{5}$, then:

$$\frac{A(\triangle_1)}{A(\triangle_2)} = \left(\frac{3}{5}\right)^2 = \frac{9}{25}$$

$$\boxed{\text{Ratio of areas} = 9 : 25}$$

(2) Find the side of a square if its diagonal is $10\sqrt{2}$ cm.

If the side of a square is a, then diagonal $= a\sqrt{2}$.

$\therefore a\sqrt{2} = 10\sqrt{2}$

$\therefore a = 10$ cm

$$\boxed{\text{Side of square} = 10 \text{ cm}}$$

(3) If the measure of minor arc is 70°, then find the measure of corresponding major arc.

$m(\text{major arc}) + m(\text{minor arc}) = 360°$

$\therefore m(\text{major arc}) = 360° - 70° = 290°$

$$\boxed{m(\text{major arc}) = 290°}$$

(4) Angle made by the line with the positive direction of X-axis is 45°, find the slope of line.

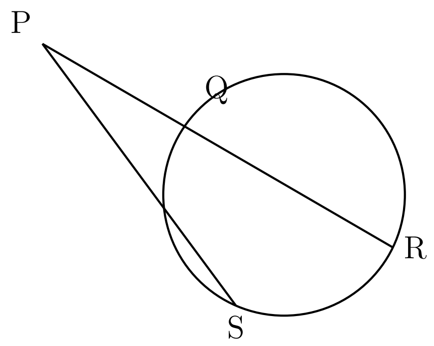
Slope $= \tan\theta = \tan 45° = 1$

$$\boxed{\text{Slope} = 1}$$

Q.2

Q.2 (A) Complete the following activities and rewrite it (any *two*): [4 Marks]

(i) In the following figure, ray PS is a tangent segment, line PR is a secant. If PQ = 3.6, QR = 6.4, find PS.



Activity:

$PS^2 = \boxed{\text{PQ}} \times PR$ (Tangent secant segment theorem)

$= PQ \times [PQ + QR]$

$= \boxed{3.6} \times [3.6 + 6.4]$

$= 3.6 \times \boxed{10}$

$\therefore PS^2 = 36$

$PS = \boxed{6}$

(ii) If $\sec\theta = \dfrac{25}{7}$**, find the value of** $\tan\theta$**.**

Solution:

$1 + \tan^2\theta = \boxed{\sec^2\theta}$

$$1 + \tan^2\theta = \left(\boxed{\frac{25}{7}}\right)^2$$

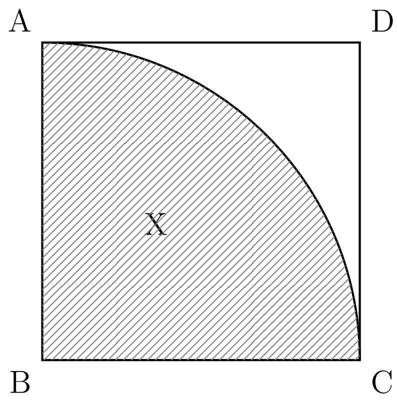
$$\tan^2\theta = \frac{625}{49} - \boxed{1}$$

$$= \frac{625 - 49}{49}$$

$$\tan^2\theta = \frac{576}{49}$$

$$\tan\theta = \boxed{\frac{24}{7}}$$

(iii) In the figure, side of square ABCD is 7 cm. With centre D and radius DA, sector D-AXC is drawn. Find the area of shaded region.



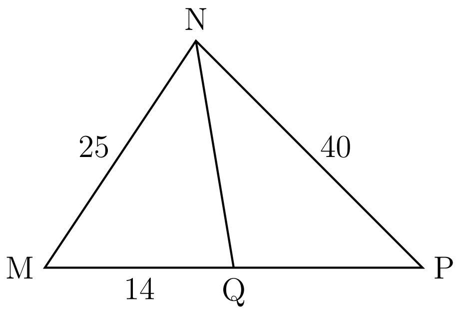
Solution:
Area of square = $(\text{side})^2 = (7)^2$
$\therefore$ Area of square = $\boxed{49}$ cm^2

If area of sector (D-AXC) = 38.5 cm^2, then
A(shaded region) = A $\boxed{\text{square}}$ − A $\boxed{\text{sector}}$
$= 49 \text{ cm}^2 - 38.5 \text{ cm}^2$
$= \boxed{10.5}$ cm^2

$$\boxed{\text{Area of shaded region} = 10.5 \text{ cm}^2}$$

Q.2 (B) Solve any *four* sub-questions from the following: [8 Marks]

(i) In $\triangle MNP$, NQ bisects $\angle MNP$. If $MN = 25$, $NP = 40$, $MQ = 14$, find QP.



By the angle bisector theorem, the angle bisector of a triangle divides the opposite side in the ratio of the adjacent sides.

$$\therefore \frac{MQ}{QP} = \frac{MN}{NP}$$
$$\therefore \frac{14}{QP} = \frac{25}{40}$$
$$\therefore QP = \frac{14 \times 40}{25} = \frac{560}{25}$$

$$\boxed{QP = 22.4}$$

(ii) In $\triangle RST$, $\angle S = 90°$, $\angle T = 30°$, $RT = 12$ cm. Find RS and ST.

In $\triangle RST$:

$\angle S = 90°$, $\angle T = 30°$

$\therefore \angle R = 180° - 90° - 30° = 60°$

Using trigonometric ratios:

$\sin T = \dfrac{RS}{RT}$

$\sin 30° = \dfrac{RS}{12}$

$\dfrac{1}{2} = \dfrac{RS}{12}$

$\therefore RS = 6$ cm

$\cos T = \dfrac{ST}{RT}$

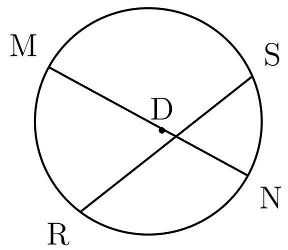
$\cos 30° = \dfrac{ST}{12}$

$\dfrac{\sqrt{3}}{2} = \dfrac{ST}{12}$

$\therefore ST = 6\sqrt{3}$ cm

$$\boxed{RS = 6 \text{ cm}, \quad ST = 6\sqrt{3} \text{ cm}}$$

(iii) In the figure, chord MN and chord RS intersect at point D. If $RD = 15$, $DS = 4$, $MD = 8$, find DN.



By the property of intersecting chords:

$MD \times DN = RD \times DS$

$\therefore 8 \times DN = 15 \times 4$

$\therefore 8 \times DN = 60$

$\therefore DN = \dfrac{60}{8}$

$$\boxed{DN = 7.5}$$

(iv) Find the coordinates of the midpoint of the line segment joining points $P(0, 6)$ and $Q(12, 20)$.

Using the midpoint formula:

$M = \left(\dfrac{x_1 + x_2}{2}, \dfrac{y_1 + y_2}{2}\right)$

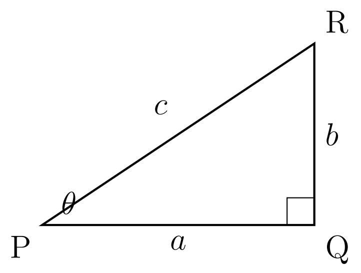
$= \left(\dfrac{0 + 12}{2}, \dfrac{6 + 20}{2}\right)$

$$= \left(\frac{12}{2}, \frac{26}{2}\right)$$

$$\boxed{M = (6, 13)}$$

(v) Prove that: $\sin^2\theta + \cos^2\theta = 1$

Proof:



In right-angled $\triangle PQR$, $\angle Q = 90°$.
Let $PQ = a$, $QR = b$, $PR = c$.

By Pythagoras theorem:
$PQ^2 + QR^2 = PR^2$
$\therefore a^2 + b^2 = c^2 \quad \ldots \text{(I)}$

Now, $\sin\theta = \dfrac{\text{Opposite side}}{\text{Hypotenuse}} = \dfrac{QR}{PR} = \dfrac{b}{c}$

$\cos\theta = \dfrac{\text{Adjacent side}}{\text{Hypotenuse}} = \dfrac{PQ}{PR} = \dfrac{a}{c}$

$\therefore \sin^2\theta + \cos^2\theta = \dfrac{b^2}{c^2} + \dfrac{a^2}{c^2} = \dfrac{a^2+b^2}{c^2}$

From (I), $a^2 + b^2 = c^2$.

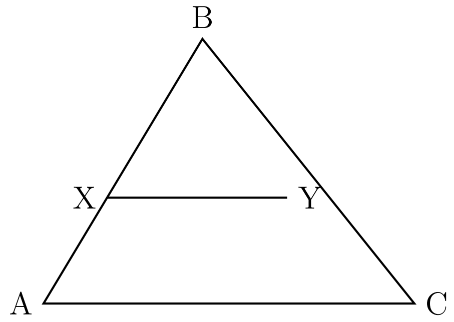
$\therefore \sin^2\theta + \cos^2\theta = \dfrac{c^2}{c^2} = 1$

$$\boxed{\sin^2\theta + \cos^2\theta = 1}$$

Q.3

Q.3 (A) Complete the following activities and rewrite it (any *one*): [3 Marks]

(i) In the figure, seg $XY \parallel$ side AC. If $2AX = 3BX$ and $XY = 9$, complete the activity to find the value of AC.



Activity:

$2AX = 3BX$

$\therefore \frac{AX}{BX} = \frac{3}{2}$

$\frac{AX + BX}{BX} = \frac{\boxed{3} + \boxed{2}}{2}$ (By componendo)

$\frac{AB}{BX} = \frac{\boxed{5}}{2}$

$\triangle BCA \sim \triangle BYX$ ...($\boxed{\text{AA}}$ test of similarity)

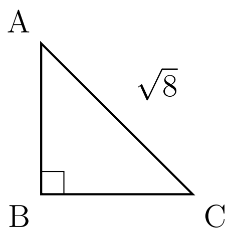
$\therefore \frac{BA}{BX} = \frac{AC}{XY}$ ...(Corresponding sides of similar triangles)

$\therefore \frac{5}{\boxed{2}} = \frac{AC}{9}$

$\therefore AC = \boxed{22.5}$

(ii) For finding AB and BC with the help of information given in the figure, complete the following activity.

In $\triangle ABC$, $AB = BC$, $\angle B = 90°$ and $AC = \sqrt{8}$.



Activity:

$AB = BC$ ...$\boxed{\text{Given}}$

$\therefore \angle BAC = \boxed{45^\circ}$

$\therefore AB = BC = \boxed{\dfrac{1}{\sqrt{2}}} \times AC$

$= \dfrac{1}{\sqrt{2}} \times \boxed{\sqrt{8}}$

$= \dfrac{1}{\sqrt{2}} \times 2 \times \boxed{\sqrt{2}}$

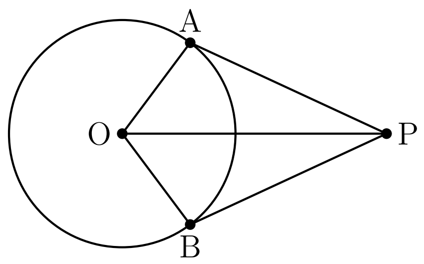
$AB = BC = \boxed{2}$

Q.3 (B) Solve any *two* sub-questions of the following: [6 Marks]

(i) Prove that: The tangent segments drawn from an external point to the circle are congruent.

Given: A circle with centre O. Point P is an external point. PA and PB are tangent segments drawn from P to the circle touching at A and B respectively.

To Prove: seg $PA \cong$ seg PB



Construction: Draw seg OA, seg OB and seg OP.

Proof:

In $\triangle OAP$ and $\triangle OBP$:

seg $OA \cong$ seg OB ...(Radii of the same circle)

$\angle OAP = \angle OBP = 90^\circ$ ...(Tangent is perpendicular to radius)

seg $OP \cong$ seg OP ...(Common side)

$\therefore \triangle OAP \cong \triangle OBP$ ...(By Hypotenuse-Side test)

$\therefore$ seg $PA \cong$ seg PB ...(c.s.c.t.)

$\boxed{\text{Hence proved.}}$

(ii) Draw a circle with centre O and radius 3.5 cm, take a point P at a distance 5.7 cm from the centre. Draw tangents to the circle from point P.

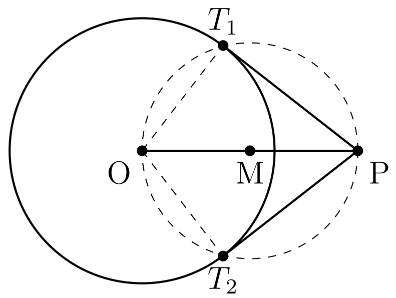
Steps of construction:

1. Draw a circle with centre O and radius 3.5 cm.
2. Take a point P outside the circle such that $OP = 5.7$ cm.
3. Draw segment OP and find its midpoint M.

4. With M as centre and OM as radius, draw a circle intersecting the first circle at points T_1 and T_2.

5. Draw lines PT_1 and PT_2.

PT_1 and PT_2 are the required tangent segments.



(iii) Find the co-ordinates of point P, if P divides the line segment joining the points $A(-1, 7)$ and $B(4, -3)$ in the ratio 2 : 3.

Using the section formula:

$$P = \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}\right)$$

Here $A(x_1, y_1) = (-1, 7)$, $B(x_2, y_2) = (4, -3)$, $m : n = 2 : 3$

$$x = \frac{2(4) + 3(-1)}{2+3} = \frac{8-3}{5} = \frac{5}{5} = 1$$

$$y = \frac{2(-3) + 3(7)}{2+3} = \frac{-6+21}{5} = \frac{15}{5} = 3$$

$$\boxed{P = (1, 3)}$$

(iv) A bucket is frustum shaped. Its height is 28 cm. Radii of circular faces are 12 cm and 15 cm. Find the capacity of the bucket in litres. $\left(\pi = \frac{22}{7}\right)$

The bucket is in the shape of a frustum of a cone.

Here $h = 28$ cm, $r_1 = 15$ cm, $r_2 = 12$ cm

$$\text{Volume of frustum} = \frac{\pi h}{3}(r_1^2 + r_2^2 + r_1 \times r_2)$$

$$= \frac{22}{7} \times \frac{28}{3} \times (15^2 + 12^2 + 15 \times 12)$$

$$= \frac{22}{7} \times \frac{28}{3} \times (225 + 144 + 180)$$

$$= \frac{22}{7} \times \frac{28}{3} \times 549$$

$$= \frac{22 \times 4}{3} \times 549$$

$$= \frac{88 \times 549}{3}$$

$$= \frac{48312}{3}$$

$$= 16104 \text{ cm}^3$$

1 litre = 1000 cm^3
$\therefore$ Capacity $= \dfrac{16104}{1000} = 16.104$ litres

$$\boxed{\text{Capacity of bucket} = 16.104 \text{ litres}}$$

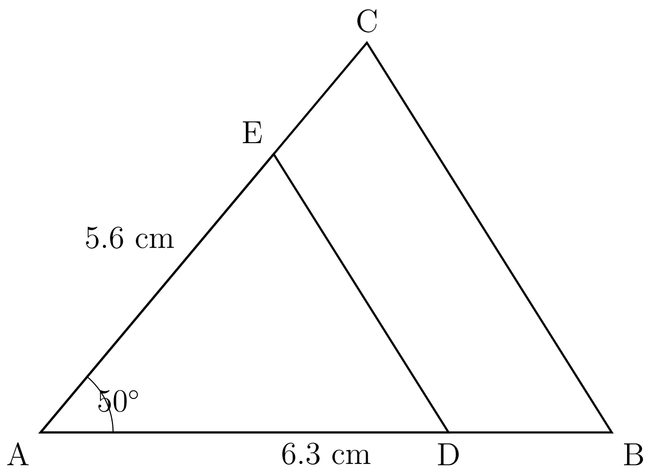
Q.4 Solve any *two* sub-questions of the following: [8 Marks]

(i) $\triangle ABC \sim \triangle ADE$. In $\triangle ABC$, $AB = 6.3$ cm, $\angle CAB = 50^\circ$, $AC = 5.6$ cm and $\dfrac{AB}{AD} = \dfrac{7}{5}$. Construct $\triangle ADE$.

$\dfrac{AB}{AD} = \dfrac{7}{5}$, so the scale factor is $\dfrac{5}{7}$ (since $\triangle ADE$ is smaller).

Steps of construction:

1. Draw $\triangle ABC$ with $AB = 6.3$ cm, $\angle CAB = 50^\circ$, and $AC = 5.6$ cm.
2. Draw ray AB. Divide segment AB in the ratio $5 : 7$.
   - Draw ray AM making an acute angle with ray AB.
   - Mark 7 equal divisions $A_1, A_2, \ldots, A_7$ on ray AM.
   - Join A_7B.
   - Through A_5, draw a line parallel to A_7B meeting AB at D.
3. Through D, draw a line parallel to side BC meeting AC at E.
4. $\triangle ADE$ is the required triangle.



(ii) Surface area of sphere and total surface area of cube are same. Show that the ratio of the volume of the sphere to that of the cube is $\sqrt{6} : \sqrt{\pi}$.

Let the radius of the sphere be r and the side of the cube be a.

Given: Surface area of sphere = Total surface area of cube

$4\pi r^2 = 6a^2$

$\therefore \dfrac{r^2}{a^2} = \dfrac{6}{4\pi} = \dfrac{3}{2\pi}$

$\therefore \dfrac{r}{a} = \sqrt{\dfrac{3}{2\pi}} \quad \ldots \text{(I)}$

To show: $\dfrac{V_{\text{sphere}}}{V_{\text{cube}}} = \dfrac{\sqrt{6}}{\sqrt{\pi}}$

$\dfrac{V_{\text{sphere}}}{V_{\text{cube}}} = \dfrac{\frac{4}{3}\pi r^3}{a^3} = \dfrac{4\pi}{3} \times \dfrac{r^3}{a^3} = \dfrac{4\pi}{3} \times \left(\dfrac{r}{a}\right)^3$

From (I): $\dfrac{r}{a} = \sqrt{\dfrac{3}{2\pi}}$

$\therefore \left(\dfrac{r}{a}\right)^3 = \left(\dfrac{3}{2\pi}\right)^{3/2} = \dfrac{3\sqrt{3}}{2\pi\sqrt{2\pi}}$

$\therefore \dfrac{V_{\text{sphere}}}{V_{\text{cube}}} = \dfrac{4\pi}{3} \times \dfrac{3\sqrt{3}}{2\pi\sqrt{2\pi}}$

$= \dfrac{4\pi \times 3\sqrt{3}}{3 \times 2\pi\sqrt{2\pi}}$

$= \dfrac{4\sqrt{3}}{2\sqrt{2\pi}}$

$= \dfrac{2\sqrt{3}}{\sqrt{2\pi}}$

$= \dfrac{2\sqrt{3}}{\sqrt{2} \cdot \sqrt{\pi}}$

$= \dfrac{2\sqrt{3}}{\sqrt{2} \cdot \sqrt{\pi}} \times \dfrac{\sqrt{2}}{\sqrt{2}}$

$= \dfrac{2\sqrt{6}}{2\sqrt{\pi}}$

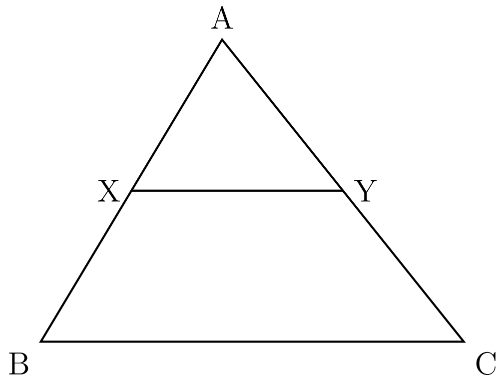
$= \dfrac{\sqrt{6}}{\sqrt{\pi}}$

$$\boxed{\dfrac{V_{\text{sphere}}}{V_{\text{cube}}} = \dfrac{\sqrt{6}}{\sqrt{\pi}}}$$

$\therefore$ The ratio of the volume of the sphere to that of the cube is $\sqrt{6} : \sqrt{\pi}$.

Hence proved.

(iii) In $\triangle ABC$, A–X–B and A–Y–C is such that segment $XY \parallel$ side BC. Segment XY divides $\triangle ABC$ in two equal areas, then find $\frac{BX}{AB}$.



Since $XY \parallel BC$, by Basic Proportionality Theorem:
$\triangle AXY \sim \triangle ABC \quad \ldots$(AA test of similarity)

$\therefore \dfrac{A(\triangle AXY)}{A(\triangle ABC)} = \dfrac{AX^2}{AB^2} \quad \ldots$(Areas of similar triangles)

Given: Segment XY divides $\triangle ABC$ into two equal areas.

$\therefore A(\triangle AXY) = \frac{1}{2}A(\triangle ABC)$

$\therefore \dfrac{A(\triangle AXY)}{A(\triangle ABC)} = \dfrac{1}{2}$

$\therefore \dfrac{AX^2}{AB^2} = \dfrac{1}{2}$

$\therefore \dfrac{AX}{AB} = \dfrac{1}{\sqrt{2}}$

Now, $BX = AB - AX$

$\therefore \dfrac{BX}{AB} = \dfrac{AB - AX}{AB} = 1 - \dfrac{AX}{AB} = 1 - \dfrac{1}{\sqrt{2}}$

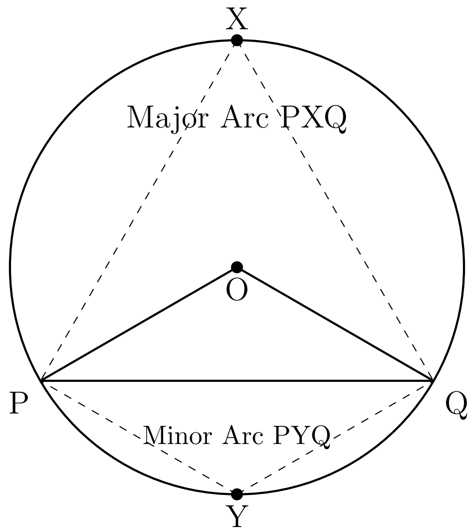
$= \dfrac{\sqrt{2}-1}{\sqrt{2}}$

$$\boxed{\frac{BX}{AB} = \frac{\sqrt{2}-1}{\sqrt{2}} = \frac{2-\sqrt{2}}{2}}$$

Q.5 Solve any *one* sub-question of the following: [3 Marks]

(i) Seg PQ is a chord of a circle. Arc PXQ is a major arc and Arc PYQ is a minor arc. $\angle POQ$ is the central angle of the circle.
(a) Draw the figure from given information.
(b) Find the measure of $\angle PXQ$, $\angle PYQ$ and add $m\angle PXQ + m\angle PYQ$.
(c) Find the measure of $\angle POQ$. Write the relation between $\angle PXQ$ and $\angle POQ$.

(a) Figure:



(b) Let $m(\text{arc PYQ}) = x^\circ$ (minor arc).
Then $m(\text{arc PXQ}) = (360 - x)^\circ$ (major arc).

$\angle PXQ$ is an inscribed angle intercepting minor arc PYQ.
$\therefore m\angle PXQ = \frac{1}{2} \times m(\text{arc PYQ}) = \frac{x}{2}$

$\angle PYQ$ is an inscribed angle intercepting major arc PXQ.
$\therefore m\angle PYQ = \frac{1}{2} \times m(\text{arc PXQ}) = \frac{360 - x}{2}$

$$m\angle PXQ + m\angle PYQ = \frac{x}{2} + \frac{360 - x}{2} = \frac{x + 360 - x}{2} = \frac{360}{2}$$

$$\boxed{m\angle PXQ + m\angle PYQ = 180^\circ}$$

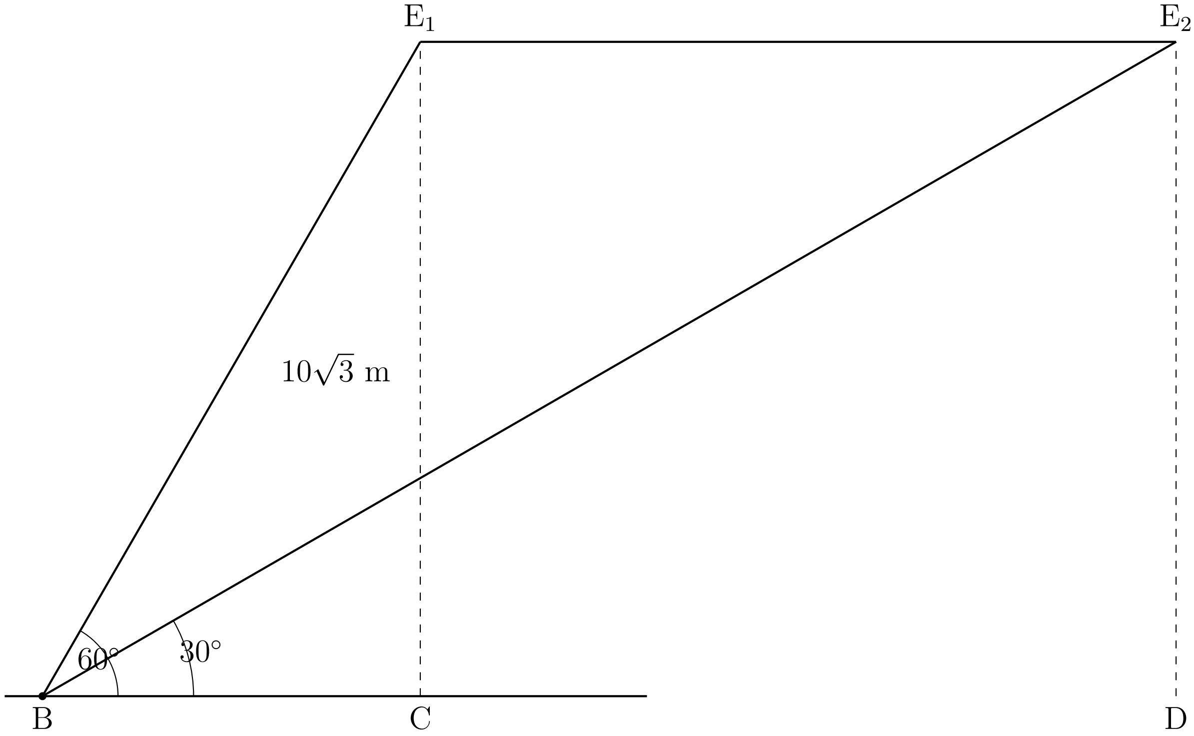
(c) $\angle POQ$ is the central angle intercepting minor arc PYQ.
$\therefore m\angle POQ = m(\text{arc PYQ}) = x^\circ$

Since $m\angle PXQ = \frac{x}{2} = \frac{m\angle POQ}{2}$

$$\boxed{m\angle POQ = 2 \times m\angle PXQ}$$

The central angle is twice the inscribed angle intercepting the same arc.

(ii) Two eagles are flying in the sky at a height of $10\sqrt{3}$ m above the ground. A boy standing at a point on the ground looks towards the eagles making angles of elevation of 60° and 30° respectively. Find the distance between the two eagles.



Let the boy stand at point B on the ground. Let E_1 and E_2 be the positions of the two eagles at height $10\sqrt{3}$ m.

Let BC and BD be the horizontal distances of E_1 and E_2 from B respectively, where C and D are the feet of the perpendiculars from E_1 and E_2.

For Eagle E_1 (angle of elevation = 60°):

$$\tan 60^\circ = \frac{E_1C}{BC} = \frac{10\sqrt{3}}{BC}$$

$$\sqrt{3} = \frac{10\sqrt{3}}{BC}$$

$$\therefore BC = \frac{10\sqrt{3}}{\sqrt{3}} = 10 \text{ m}$$

For Eagle E_2 (angle of elevation = 30°):

$$\tan 30^\circ = \frac{E_2D}{BD} = \frac{10\sqrt{3}}{BD}$$

$$\frac{1}{\sqrt{3}} = \frac{10\sqrt{3}}{BD}$$

$$\therefore BD = 10\sqrt{3} \times \sqrt{3} = 30 \text{ m}$$

Since both eagles are at the same height, the distance between them is:
Distance $= BD - BC = 30 - 10 = 20$ m

$$\boxed{\text{Distance between the two eagles} = 20 \text{ m}}$$

— End of Solutions —